

How to change the net name of a wire segment or via ?

```
editSelect -nets VDD1
editSelectVia -nets VDD1
editChangeNet -to VDD2
```

How to change the wire width of a special net in Innovus ?

```
editSelect -nets <netname>
editChangeWidth -width_horizontal <value> -width_vertical <value>
```

you may consider using dbSelectObj when selecting special nets of a particular width:

```
dbSelectObj [dbGet -p [dbGet -p top.nets.name <net_Name>].sWires.width <width> ]
```

For example, the following will select the wire segments of the VSS net that has a width of 1.6:

```
dbSelectObj [dbGet -p [dbGet -p top.nets.name VSS ].sWires.width 1.6 ]
```

Look for all options of Below writeFPlanScript commnd

```
writeFPlanScript
```

floorplan boundary

```
set ret [catch {floorplan -noResize -noSnapToGrid -b 0.000 0.000 \
1000.000 498.000 0.000 0.000 1000.000 498.000 5.016 5.010 594.984 492.990 }]
```

```
if {$ret != 0} {
    floorplan -noSnapToGrid -b 0.000 0.000 1000.000 498.000 0.000 0.000 \
1000.000 498.000 5.016 5.010 594.984 492.990
}
```

Blocks placemnet

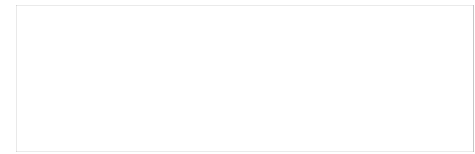
```
unplaceAllBlocks
deleteHaloFromBlock -allBlock
placeInstance Memory_1 377.696 464.910 R180
addHaloToBlock 2 2 2 2 Memory_1
setInstancePlacementStatus -status fixed -name Memory_1
```

Placement obstruction

```
deletePlaceBlockage -all
createPlaceBlockage -type hard -name defScreenName -box {218.9 280.7 315.95 382.53}
createPlaceBlockage -type soft -name defScreenName -box {218.9 280.7 315.95 382.53}
```

Standard Cell Row

```
deleteRow -all
set rdagFPlanRowSpaceType 0
set rdagFPlanRowSpacing 0
createRow -site HDCORE_VDD -area 5.016 5.010 594.966 5.490 -noCheck
```



Routing obstruction

```
deleteRouteBlk -all
createRouteBlk -name defLayerBlkName -layer M2 -boxList { 69.03 406.6 124.1 457.31700 }
createRouteBlk -name defLayerBlkName -layer {M2 M1 M3} -boxList { 69.03 406.6 124.1
457.31 }
```

Pin blockage

```
deletePinBlkg -all
```

Creating Instance group

```
#the group SCH is belong to power domain : SCH,
#Please use it carefully, for example don't delete it.
deleteInstGroup VI_H0
createInstGroup VI_H0
addInstToInstGroup VI_H0 snps_SCH__dft_outputs_iso_low_snps_ib_scan_out*
createRegion VI_H0 214.491000 521.520000 260.490000 526.080000
```

Pin section

```
setPinAssignMode -pinEditInBatch true
editPin -pin clk -layer M7 -assign {189.924 528.96} -pinWidth 0.076 \
-pinDepth 0.395 -fixOverlap false -side N -fixedPin
editPin -pin proc_mrs -layer M7 -assign {190.152 528.96} -pinWidth \
0.076 -pinDepth 0.395 -fixOverlap false -side N -fixedPin
setPinAssignMode -pinEditInBatch false
```

Global Net Connection

```
clearGlobalNets
globalNetConnect vdd_modem -pin {vdd_sub} -type pgpin -all
globalNetConnect vdd_modem -pin {vddaon} -type pgpin -all
globalNetConnect vdd_modem -pin {vdd_ext} -type pgpin -all

globalNetConnect vdd_sch_int -type tiehi -all
globalNetConnect vdd_sch_int -pin {vddfx} -type pgpin -all
globalNetConnect vdd_sch_int -pin {vddx} -type pgpin -all

globalNetConnect vssx_0 -type tielo -all
globalNetConnect vssx_0 -pin {vssfx} -type pgpin -all
globalNetConnect vssx_0 -pin {vssx} -type pgpin -all
globalNetConnect vdd_sch_int -pin {vddmx} -singleInst instanceName -type pgpin

applyGlobalNets
```

Usage: editSelect [-help] [-area {x1 y1 x2 y2}] [-direction {H V 45 135}] \
[-layer {layer | list_of_layers}]
[-net {net | list_of_nets}]
[-shape {RING STRIPE FOLLOWPIN IOWIRE COREWIRE BLOCKWIRE \
PADRING BLOCKRING FILLWIRE FILLWIREOPC DRCFILL None}]
[-shield {0 1}] [-status {COVER FIXED NOSHIELD ROUTED SHIELD}]\
[-subclass <string>]
[-type {Regular Special Patch}] [-use {CLOCK POWER SIGNAL}]\
[-wires_only {0 1}] [-with_edit_flag {0 1}]

```
editSelect -type Special -net vssx_0 -layer M11
editSelect -type Special -net vdd_modem -layer M11
```

```
innovus > dbGet top.fplan.bndrys.??
boxes: {920.7 340.48 1114.74 511.84}
hInst: 0x18eabc20
objType: bndry
type: guide
boxes: {1057.98 582.4 1339.14 829.36}
hInst: 0x18eabcb0
objType: bndry
type: cluster
```

#To change the constraint type of a group

For example to change group from region to guide -

```
innovus > dbGet -p top.fplan.groups.conType
region
innovus > dbSet [dbGet -p top.fplan.groups.name group_name].conType guide
```

To change constraint type of all groups:

```
dbSet top.fplan.groups.conType none
```

##To get/filter pointers of all the fences

```
dbForEachFPlanConstraint [dbHeadFPlan] constPtr {
    set ctype [dbConstraintType $constPtr]
    puts "$ctype"
    if {$ctype == "dbcFence"} {
        puts "$constPtr"
    }
}
```

To select/delete an object in the design

The below command can select rows based on pointer of the rows which are obtained through matching pattern, then delete those rows.

```
dbSelectObj [dbGet -p2 top.fplan.rows.site.name core10]
```

```
dbDeleteObj [dbGet -p2 top.fplan.rows.site.name core10]
```

To report information on placement blockages

```
innovus > dbGet top.fplan.pBlks.??
```

```
attr: undefined
```

```
boxes: {{692.34 1025.92 933.9 1081.36}}
```

```
density: 0
```

```
inst: 0x0
```

```
Name: defScreenName
```

```
objType: pBlkg
```

```
shapes: 0x1935ca20
```

```
type: partial
```

Report halo sizes on each side

```
dbGet [dbGet -p2 top.insts.cell.subClass block*].pHaloTop
```

```
dbGet [dbGet -p2 top.insts.cell.subClass block*].pHaloBot
```

```
dbGet [dbGet -p2 top.insts.cell.subClass block*].pHaloLeft
```

```
dbGet [dbGet -p2 top.insts.cell.subClass block*].pHaloRight
```

Report size and top/bottom layers of block routing halos

```
dbGet [dbGet -p2 top.insts.cell.subClass block*].rHaloSideSize
```

```
dbGet [dbGet -p2 top.insts.cell.subClass block*].rHaloBotLayer.name
```

```
dbGet [dbGet -p2 top.insts.cell.subClass block*].rHaloTopLayer.name
```

Adding/Selecting/Removing blockages

```
createRouteBlockage -name route_blockage1 -layer M1
```

```
selectRouteBlk -layer M1 route_blockage1
```

```
deleteRouteBlk -layer M1 -name route_blockage1
```

```
innovus > dbget selected.??
```

```
area: 916.99776
```

```
attr: default
```

```
boxes: {{0.0 109.44 0.684 1450.08}}
```

```
density: 0
```

```
designRuleWidth: 0.0
```

```
inst: 0x0
```

```
isExceptPGNet: 1
```

```
isPGNetOnly: 0
```

```
isPushdown: 0
```

```
layer: 0x2adb25c29000
```

```
name: bdry_r1
```

```
objType: rBlkg
```

shapes: 0x2adc2672ae00

spacing: 0.

Note: The isPushDown attribute can be used to add blockage one level down the design.

dbSet selected.isPushDown 1

A new LEF MASTER SLICE layer TRIMMETAL is introduced in implementation flow. Trim (cut) metal is used to shrink the end-to-end spacing, enclosure and MAR, which should be always sandwiched by 2 wire line-ends
createRouteBlk -trimMetalLayer TL1 TL2

##Reporting types of Physical only cells used in design

dbget [dbget top.physInsts.isPhysOnly 1 -p].cell.name -u

For example : Endcap cells ,Filler cells

Report all leaf cells through the complete hierarchy

dbGet top.insts

Return pointer for all of the leaf cell and hierarchical instances through the complete hierarchy

dbGet top.hInst.alltreeinsts

Return pointer of all the hierarchical instances through the complete hierarchy

dbGet -p top.hInst.alltreeinsts.objType hInst

To report row names for IO rows and Core rows

dbget [dbget -p2 dbget top.fPlan.rows.site.class pad].name

dbget [dbget -p2 dbget top.fPlan.rows.site.class core].name

##To select rows in the core area

select_row -site site0A10

To determine the orientation of the first (bottom) row use the following

dbGet top.fplan.flipRows

first

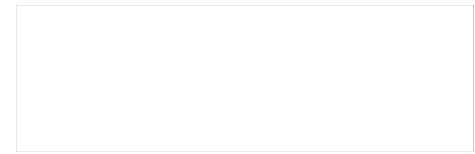
Possible return values are:

none = no flipping; (so, first row has orientation = N)

first = first row is flipped, other rows alternate; (i.e 1st row orientation = S)

second = first row is not flipped, other rows alternate

(i.e. 1st row has N orientation)



To edit nets

editSelect and selectNet commands can be used to select nets .

The difference between two is editSelect will select wire. selectNet will select the net. If you want to delete wire portion of the net please do the following steps:

```
editSelect -net <net_name>
editDelete -selected
```

To edit Pins

editPin

Modifies properties of pins, such as pin spreading, pin location, pin width and depth, spacing, snap-to location, and status. The editPin command can be used

Also PinEditor form can be used which has the equivalent capabilities .

PinEditor form can be accessed from Edit--> PinEditor

##Report pin details of all Pin Groups

```
proc reportPinGroupPins {fileName} {
    set fp [open $fileName w]
    set allPinGrps [dbget top.fplan.pinGroups]
    if { $allPinGrps != "0x0" } {
        foreach pinGroup $allPinGrps {
            puts $fp "Below are pins from Pin Group [dbget $pinGroup.name] ---> "
            dbForEachPinGroupPin $pinGroup pin {
                set pinName [dbget $pin.name]
                puts $fp "$pinName"
            }
        }
    } else {
        Puts "No Pin Groups Loaded"
    }
}
close $fp
}
```

TIPS RELATED TO MSV DESIGNS

Report group and power domain information

```
dbGet top.fplan.groups.??
```

```
dbGet top.fplan.groups.pd.??
```

#To Check Site of Power Domain

```
innovus > dbPowerDomainTechSite [dbGetPowerDomainByName VDD1]
```

```
0x73925620
```

```
innovus > dbGet 0x73925620.name
```

```
Hv_core
```

To get boxes of PowerDomain1:

```
innovus > dbget [dbget -p top.fPlan.groups.pd.name PowerDomain1].boxes
```

```
{{15.5 30.0} {150.2 200.4}}
```

To select a power domain

```
innovus > dbget top.pds.name
VDD_SW dtmf_io/dtmf_core/vdd_sw _internal_vddio_vdd_sw
innovus > selectGroup VDD_SW
innovus > dbget selected.??
area: 192.62
boxes: {{331.284 199.2 358.188 210.24} {331.284 418.08 358.188 423.84} {331.284 841.44
358.188 847.2} {331.284 1027.68 358.188 1038.72} {414.96 1553.52 439.812 1757.52}
{414.96 851.76 439.812 1055.76} {414.96 257.52 439.812 461.52}}
conType: fence
density: 0.948012034038958
members: 0x2afaf8a68110 0x2afaf8a79550 0x2afaf8a79660
name: VDDA_SW
objType: group
parent: 0x0
pd: 0x2afb207d1880
```

To check if PG nets on top are connected correctly to their macro cellpins

```
innovus > dbget top.insts.cell.baseClass block -p
0x2b8eb981e5b0
innovus > dbget 0x2b8eb981e5b0.pgInstTerms.name
vddsw vddao vss
innovus > dbget 0x2b8eb981e5b0.pgInstTerms.net.name
vddsw_1 vddao_1 vss_1
```

To select Isolation/LS cells

```
innovus > foreach LS_inst [dbget top.insts.isLevelShifter 1 -p] {
  selectInst [dbget $LS_inst.name]
}
innovus > foreach ISO_inst [dbget top.insts.isIsolation 1 -p ] {
  selectInst [dbget $ISO_inst.name]
}
```

To check supplies of power domain

```
reportPowerDomain -powerDomain VDD_CX -pgNet
Frequently used Commands at Floorplan step
writeFPlanScript
finishFloorplan
defOutBySection
```

defIn , defOut , saveFlan , loadFPlan , saveDesign , restoreDesign ,
freeDesign ,cutRow , createRow , initCoreRow, addEndcap,
verifyEndcap, addWellTap , verifyWelTap , verifyDrc

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